ShareSphere

Access More. Waste Less.

Verified sharing infrastructure for idle laboratory, workshop, camera, computing and classroom resources.

1. Project Title and Team Details

Item	Details
Project title	ShareSphere - Access More. Waste Less.
Tagline	Verified sharing infrastructure
Live application	https://classic-share-sphere-hub.base44.app
Team size	1 (individual submission)
Team member	Naveena Sandhi - Team Member 1 / Developer
Build platform	Base44 (AI-assisted app builder)
Report date	29 October 2026

2. Selected Use Case

Shared resource access and fair allocation of underused equipment. ShareSphere is a platform where colleges, laboratories, makerspaces and community organisations list equipment and spaces that sit idle for most of the year, and where verified students, researchers and startups can discover, request and book them under clear eligibility, training and supervision rules.

3. Problem Statement and Why the Problem Matters

Expensive equipment such as confocal microscopes, laser cutters, CNC routers, drones and GPU workstations is typically owned by one department or organisation and used for only a fraction of the available hours. At the same time, students, early-stage researchers and startups cannot afford to buy it. Today, sharing is arranged informally through emails, phone calls and spreadsheets, which causes several problems:

- **Waste:** capable equipment sits idle while nearby people go without access.
- **Safety and liability:** owners cannot easily confirm that a borrower is who they claim to be, or is trained to operate hazardous machinery.
- **Unfair allocation:** scarce equipment goes to whoever asks first or knows the owner, with no visible or explainable policy.
- **No accountability:** damage, late returns and scheduling conflicts are handled ad hoc, with no evidence trail and a risk of arbitrary penalties.

The problem matters because access to research and fabrication equipment directly affects learning outcomes, innovation and cost of entry for young researchers and startups. Better utilisation also reduces the need for duplicate purchases and lowers the waste of idle capital.

4. Target Users and Resource Owners

Role	Who they are	What they do on ShareSphere
User	Students, researchers, startups, community members	Discover resources, request bookings, join waiting lists, report incidents, submit verification
Resource Owner	Labs, departments, makerspaces, individuals who own equipment	List resources, set rules (training, supervision, priority policy, limits), approve or reject bookings, confirm returns
Institution Manager	Staff responsible for an organisation	Review verification evidence, record training completions, watch for over-use, resolve incidents with a recorded outcome
Administrator	Platform operator	Manage users and roles, organisations, categories, fair-use limits, demo controls and maintenance notices

Example resource owners in the demo workspace are a college (Shakti Institute of Technology, Bengaluru) and a makerspace (OpenForge Makerspace, Pune); example users include students and an independent startup.

5. Proposed Solution

ShareSphere is a web platform that turns idle capacity into **safe, accountable and fairly allocated access**. Every listing shows its eligibility, training and supervision requirements, price and deposit up front. Access decisions rest on recorded evidence (verification and training records) rather than assumptions. Contested slots are managed with an explainable waiting list, and every important action is written to an append-only audit trail. Incidents are reviewed by a human manager, and the platform never automatically charges a deposit or suspends an account.

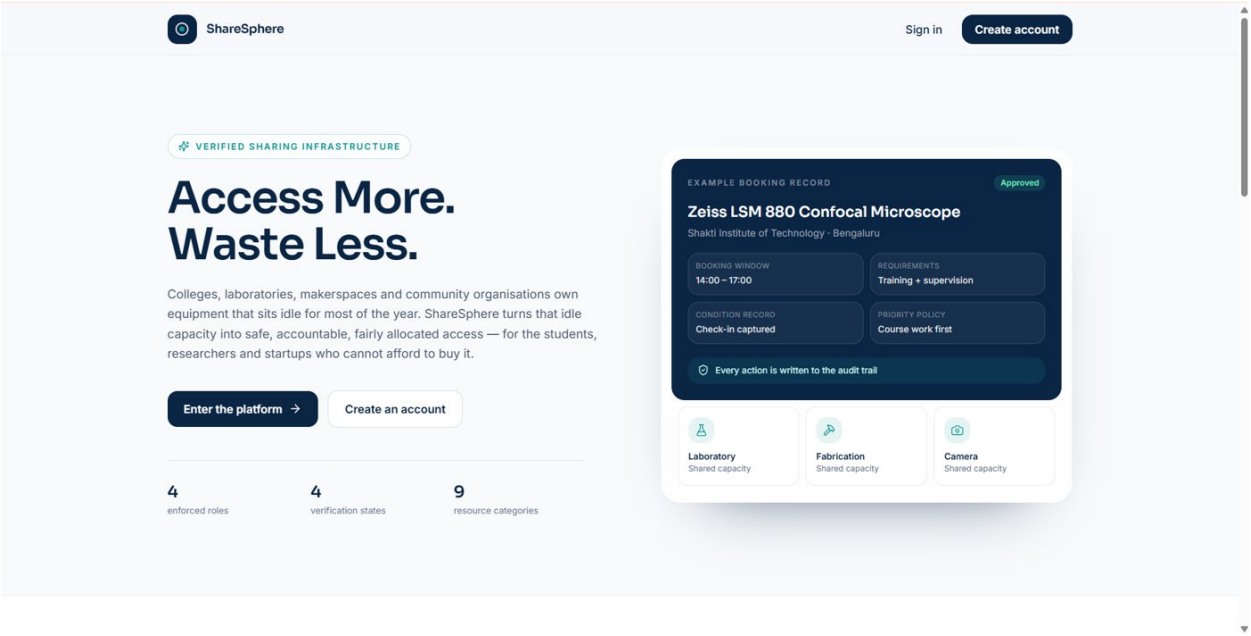


Figure 1. ShareSphere landing page with an example booking record.

6. Complete Product Workflow

#	Stage	What happens
1	Sign up and sign in	A person creates an account or signs in (email or Google). A demo workspace with fictional data lets evaluators explore every role through a "View as" switcher.
2	Get verified	The user submits their declared membership (for example student, organisation, ID reference and a document). A manager reviews the evidence, approves or rejects it with notes shown to the applicant, and the status changes (Unverified, Pending Review, Verified, or rejected) with a notification.
3	Discover resources	The user browses 12 listings across 9 categories using search, category chips, sorting and filters. "Smart match" ranks listings by purpose and budget using deterministic rules (keyword overlap, cost, availability, and whether the user's recorded training and verification satisfy the listing).
4	Check requirements	Each card and detail page shows Available status, price per booking, deposit, maximum duration, and badges such as Training required, Supervised and Verified members.
5	Request a booking	The user picks a slot. The system checks fair-use limits and eligibility, and the request enters Pending Review with the resource owner.
6	Waiting list (if the slot is taken)	The user joins a visible queue. Priority factors are recorded and shown (for example Verified member +15, no use of the resource in the last 30 days +20, owner policy such as course work first). When a booking is cancelled or rejected, the earliest entry is marked "slot offered" and notified.
7	Owner decision	The owner approves or rejects, with a reason. Approval conditions can be attached (for example induction pending, first 15 minutes supervised).
8	Use and check-in	At pickup or start, the condition record is captured. The dashboard shows what is "in your hands" and what is overdue.
9	Return and inspection	The owner confirms the return after inspection (for example "Returned clean"). Overdue items show a banner prompting return or an extension request.
10	Incidents	Any user can report damage, a late return or a scheduling conflict with severity. Reports stay in Submitted / Under Review until a manager records a finding; the outcome is shown on the report.
11	Audit and oversight	Every action (booking approved, verification approved, incident submitted or resolved, return confirmed) is recorded with actor, action, record, timestamp and reason. Administrators adjust fair-use limits and review platform totals.

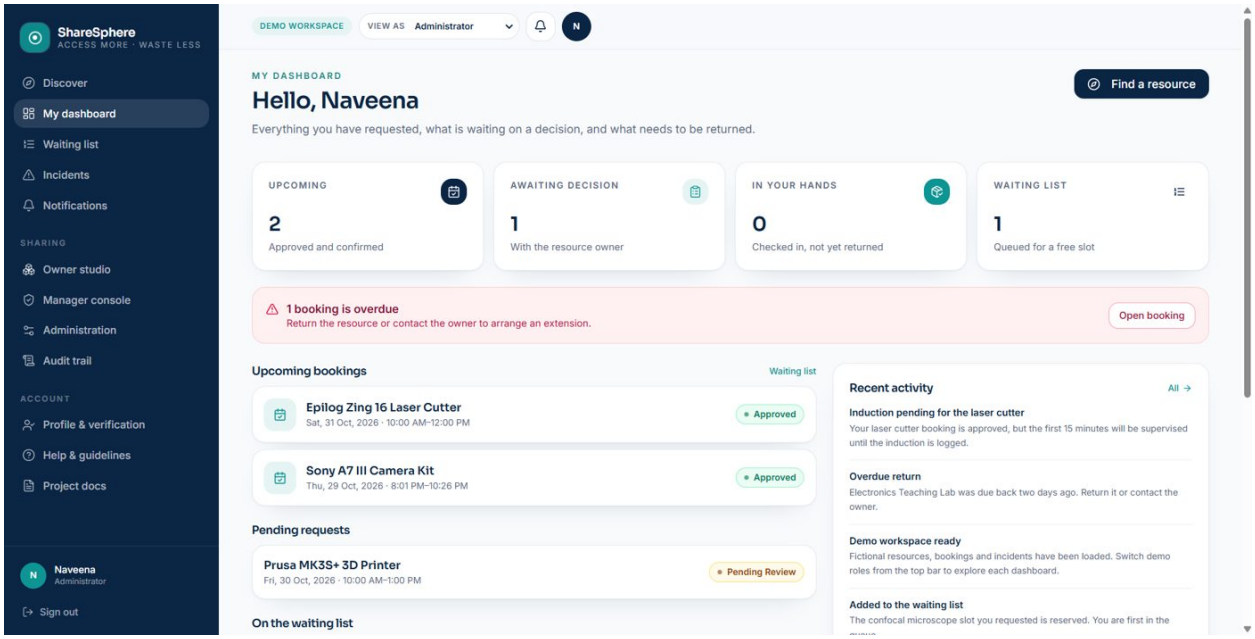


Figure 2. User dashboard: upcoming bookings, pending requests, overdue alert, waiting list and recent activity.

7. Features Developed

Discovery and listing

- Resource catalogue with search, 9 category filters (Laboratory, Classroom, Meeting Room, Computer, Camera, Fabrication Tool, Testing Device, Workshop Equipment, Other), sorting and filters.
- Smart match: purpose and budget based ranking that is deterministic and explainable, and never assumes the user is qualified.
- Resource detail, listing form for owners ("List a resource"), and Owner studio.

Booking and allocation

- Booking request, booking detail, reschedule panel, approval workflow and conditions.
- Check-in condition capture and return confirmation with inspection result.
- Overdue detection and alerts.
- Fair waiting list with visible queue positions, recorded priority factors and "slot offered" notifications; no hidden scoring and no automatic penalties.

Trust, safety and governance

- Four enforced roles and four verification states, with manager verification queue and reviewer notes.
- Training records that unlock equipment requiring them.
- Incident reporting and manager review with a recorded decision and outcome.
- Append-only audit trail with record-type filter and search.
- Administration console: people and roles, organisations, categories, fair-use limits (active bookings per user, requests per 30 days, high-demand threshold), demo mode and maintenance notice.
- In-app notifications, Help and guidelines, and Project docs pages.

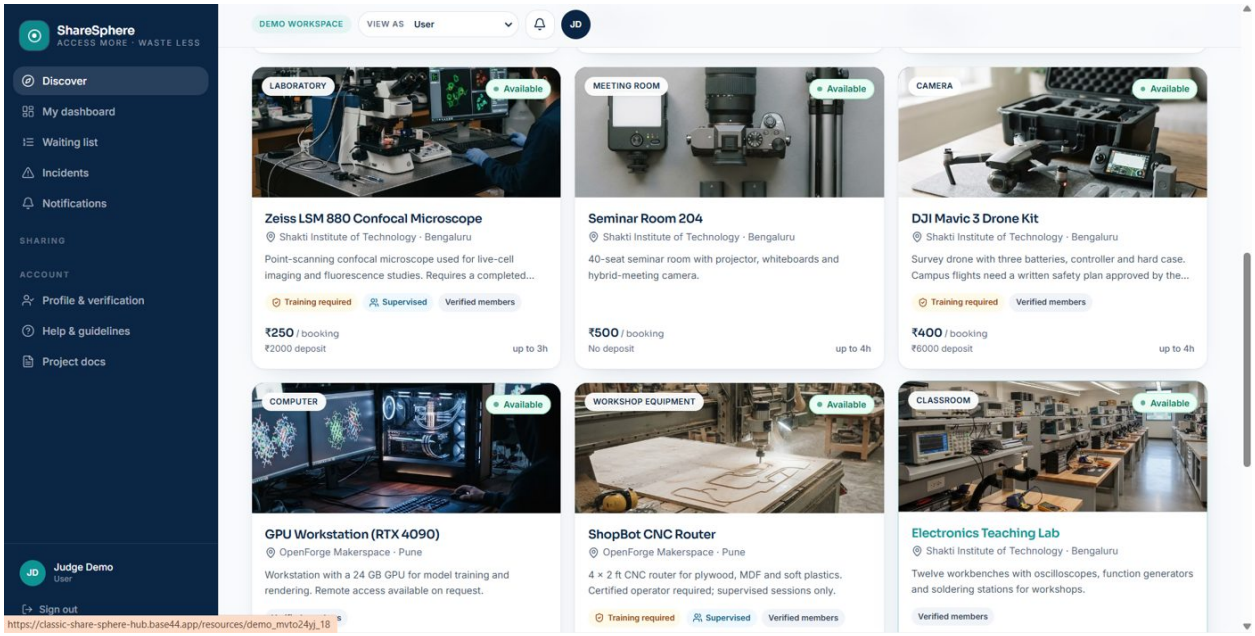


Figure 3. Discover page showing eligibility badges, price, deposit and maximum duration on every listing.

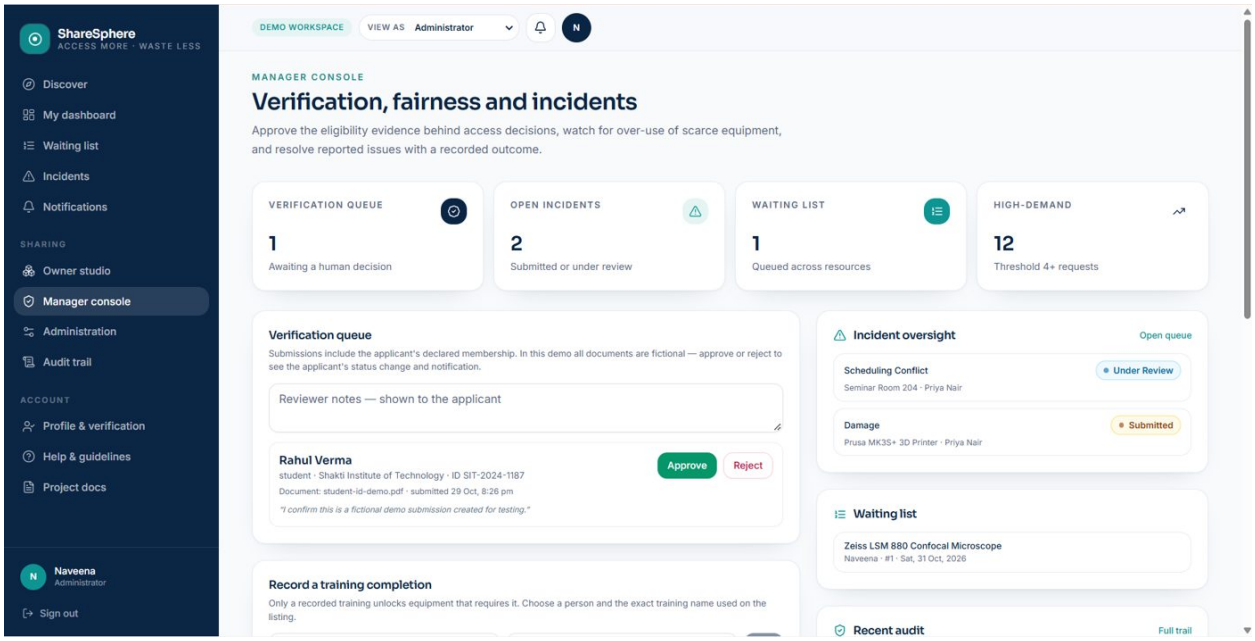


Figure 4. Manager console: verification queue, open incidents, waiting list and high-demand monitoring.

8. Technology Stack and Architecture

Layer	Technology
Platform	Base44 (hosted app builder and backend-as-a-service)
Front end	React with Vite (build tool and dev server), @vitejs/plugin-react
Styling and UI	Tailwind CSS with PostCSS; shadcn/ui-style component library (accordion, dialog, sheet, table, tabs, toast, calendar, chart and more)
Data layer	Base44 entities defined as JSONC schemas: AuditLog, Booking, ConditionReport, Incident, Notification, Organisation, PlatformSetting, Profile, Resource, TrainingRecord, User, VerificationRequest, WaitlistEntry
SDK / API client	@base44/sdk through src/api/base44Client.js

Layer	Technology
Auth	Base44 authentication with role-aware ProtectedRoute, login, register, forgot password and OAuth consent pages
Hosting	Base44 hosted deployment

Architecture

The application is a single-page React app. **src/pages** holds the screens (Landing, Login, Register, Discover, ResourceDetail, ResourceForm, BookingRequest, BookingDetail, Dashboard, Waitlist, Incidents, Notifications, OwnerDashboard, ManagerDashboard, AdminDashboard, Audit, Profile, Help, Documentation). **src/components** contains layout (AppShell), shared pieces (PageHeader, StatCard, StatusBadge, EmptyState, CategoryIcon), resource and booking components (ResourceCard, ReschedulePanel) and the UI primitives. Pages read and write the thirteen entities through the Base44 client; role checks decide which routes and actions are shown, and the AuditLog entity is written on each significant action.

9. AI Tools, APIs, Frameworks and External Services

Item	Use in this project
Base44 AI app builder	Used by the developer to scaffold, generate and iterate on the application code, entities and UI during development.
Base44 platform services	Authentication, database/entities, hosting and the SDK API used by the app at runtime.
Smart match	Implemented as deterministic rules (keyword overlap, cost, availability, eligibility check). It does **not** call a large language model at runtime.
Google sign-in	Offered on the sign-in flow through the platform's OAuth support.
Open-source frameworks	React, Vite, Tailwind CSS, shadcn/ui, Lucide icons (see Section 14).

10. Privacy, Security and Fairness

Privacy

- The demo workspace uses only fictional people, organisations, documents and bookings (emails use a non-routable demo.invalid domain). No real personal data is required to evaluate the project.
- Verification documents are visible only to authorised managers; reviewer notes are shown to the applicant.
- Only the information needed for access decisions is collected (membership type, institution, ID reference, evidence document).

Security

- Role-based access control with four roles; roles are stored per profile so one person can be both owner and manager, and pages are protected by a ProtectedRoute check.
- Sensitive actions (approving verification, resolving incidents, changing limits) require an authorised role.
- Append-only audit trail: ordinary users cannot edit or delete entries.
- Authentication handled by the Base44 platform rather than custom password storage.

Fairness

- Eligibility, training and supervision requirements are shown on every listing before a request is made.

- Waiting-list priority is explainable: each entry records the factors behind its score, which are visible to the person waiting and to the owner. "No hidden scoring is used."
- First in line is offered a freed slot first.
- No automatic penalties: a late return or incident report never demotes, charges or suspends anyone until a human manager reviews the evidence and records a finding.
- Fair-use limits (active bookings, requests per 30 days) and a high-demand threshold stop one user from monopolising scarce equipment.
- Smart match never assumes a user is qualified; it checks recorded training and verification.

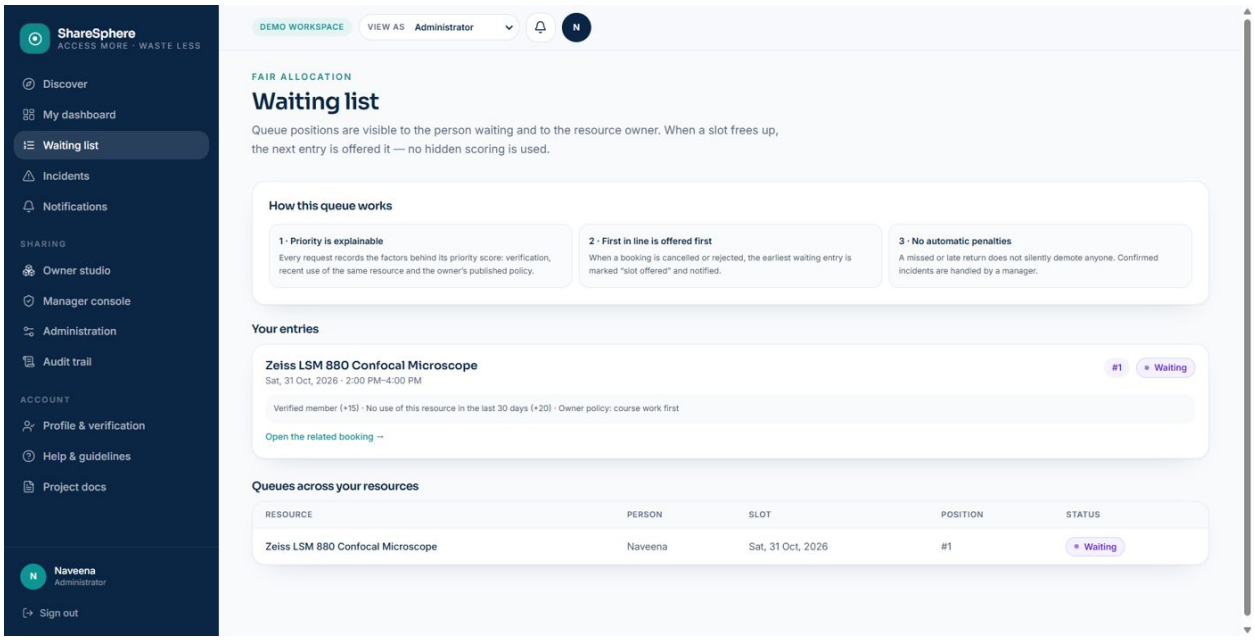


Figure 5. Waiting list showing how the queue works and the recorded priority factors.

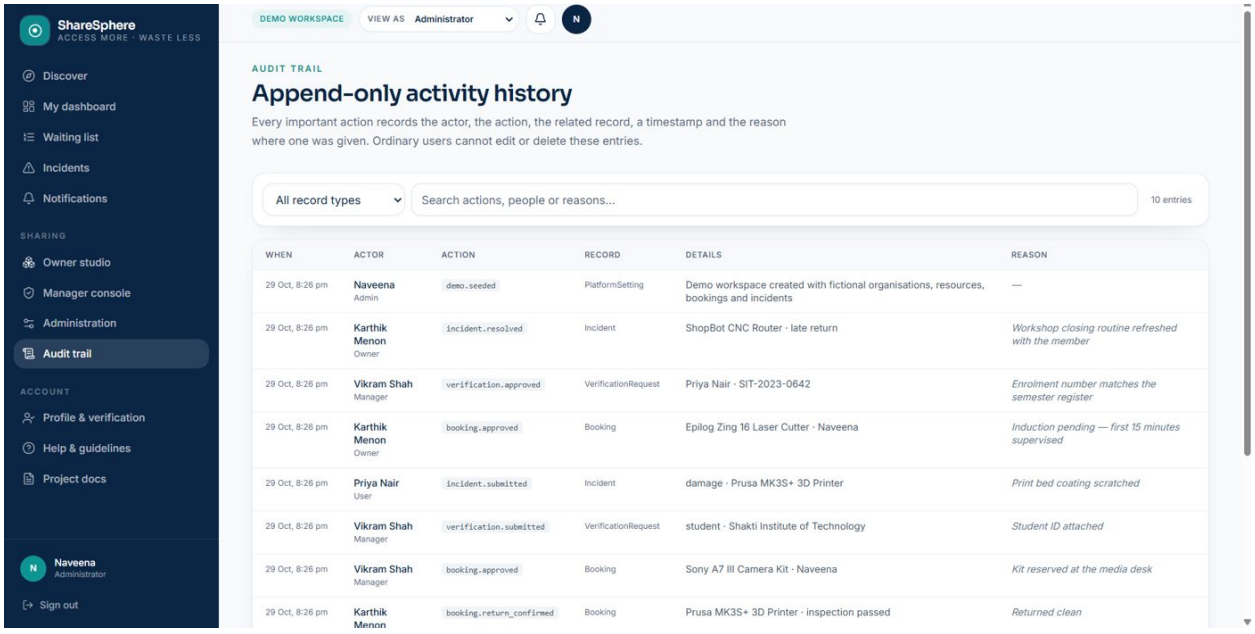


Figure 6. Append-only audit trail with actor, action, record, details and reason.

11. Known Limitations

- The project runs as a demo workspace with fictional data; it has not been piloted with real institutions.

- Verification is manual and depends on a human reviewer; there is no automated document or identity checking.
- Prices and deposits are shown on listings, but no real payment gateway is integrated, and deposits are not collected or refunded by the platform.
- Smart match is keyword and rule based, so it can miss relevant resources described with different vocabulary.
- Training completion is recorded by a manager; there is no built-in training content or assessment.
- The built-in role switcher is for demonstration only and would be removed in production.
- The current Base44 plan does not allow direct code editing in the builder, which limited fine-grained customisation to what the builder supports.
- Notifications are in-app; email, SMS or WhatsApp delivery is not yet configured.
- No formal accessibility audit, load testing or penetration test has been done.

12. Future Improvements

- Integrate a payment gateway (UPI/cards) for booking fees and refundable deposits.
- Email, SMS and calendar (Google/Outlook) integration for confirmations, reminders and slot offers.
- Optional semantic search for Smart match, with the same explainable ranking shown to the user.
- Automated identity and institutional-email verification to reduce manual review.
- Built-in short training modules and quizzes that automatically create training records.
- QR-code check-in and check-out with photo evidence for condition records.
- Usage analytics and a utilisation dashboard for owners to quantify idle time recovered.
- Multilingual interface, accessibility audit and a mobile app.
- Pilot with one college and one makerspace to refine policies and priority rules.

13. Team Members and Contributions

Name	Role	Contribution
Naveena Sandhi	Team Member 1 / Developer	Project planning, Base44 application development, feature integration, testing, and documentation.

14. Templates, Libraries and Pre-existing Code Disclosure

No code, design or content from a previous project was reused. The following third-party tools were used:

Item	Nature	Notes
Base44 platform and app scaffold	Platform / project template	The project started from the standard Base44 React project scaffold (vite.config.js with the @base44/vite-plugin, base44Client.js, config and entity files, AGENTS.md, CLAUDE.md, README.md). Application logic, entities, pages and workflows were then generated and refined by the developer using Base44.
React, Vite, @vitejs/plugin-react	Open-source libraries	Front-end framework and build tooling.

Item	Nature	Notes
Tailwind CSS, PostCSS	Open-source libraries	Utility-first styling.
shadcn/ui-style components (Radix primitives)	Open-source component library	Provided in src/components/ui (accordion, dialog, table, tabs, toast and others), used unmodified or lightly styled.
Lucide icons	Open-source icon set	Interface icons.
Listing photographs	Illustrative imagery	Used to illustrate equipment categories in the demo catalogue; they do not represent real listings.
Fictional demo data	Created for this project	All organisations, people, bookings and incidents are fictional.

Live application: https://classic-share-sphere-hub.base44.app